SWE 4847 **Security Management** **Credit: 3.00**

Information Security Management System Principles and concepts, Information Security Policy, The Integrated approach to security management, Asset Management, Strategic information security planning and management, Information security risk assessment and management framework, Information security management system Requirements, model, process and continual improvement following ISO 27001, Information security management system audit and assurance, Disaster Recovery and Business Continuity, Legal aspect of information security.

Recommended Texts:

1. M. E. Whitman and H. J. Mattord, *Management of Information Security*, 6th ed. Cengage Learning, 2018
2. E. Humphreys, "Information security management system standards," *Datenschutz und Datensicherheit - DuD*, vol. 35, no. 1, pp. 7–11, 2011. DOI: 10.1007/s11623-011-0004-3 [Online]. Available: https://link.springer.com/article/10.1007/s11623-011-0004-3

SWE 4837 **Advanced Game Development** **Credit: 3.00**

Advanced Scripting: Blueprints, Construction Scripts, Event Graphs, Material Graphs; Level Design: Environments & Terrain, Open World Tools, Encounters; Audio: Sound Cues, Ambient Sound Actors, Distance Model Attenuation; Animation: Rag-doll physics, Skeleton Retargeting, Inverse Kinematics, State Machines, Cut-Scenes with Matinee; 2D Games: Spritesheets, Flipbooks, Graphical User Interfaces; Networking: Client-Server Models, Replication; Artificial Intelligence: Pathfinding, Behaviour Trees, Rule-based/Needs-based AI; Interaction: Control & Touch Devices, Depth Sensors, Virtual/Augmented Reality.

Recommended Texts:

1. S. Madhav, *Game Programming Algorithms and Techniques: A Platform-Agnostic Approach*, 1st ed. Addison-Wesley Professional, 2013
2. T. Miller and D. Johnson, *XNA Game Studio 4.0 Programming: Developing for Windowső Phone 7 and Xbox 360*, 1st ed. Addison-Wesley Professional, 2010

SWE 4832 **OS Optimization and Real Time OS Lab** **Credit: 0.75**

Sessional works based on SWE 4831.

SWE 4834 **UI/UX Design Lab** **Credit: 0.75**

Sessional works based on SWE 4833.

SWE 4848 **Security Management Lab** **Credit: 0.75**

Sessional works based on SWE 4847.

SWE 4838 **Advanced Game Development Lab** **Credit: 0.75**

Sessional works based on SWE 4837.

Elective 8-II (Open)

CSE 4841 **Introduction to Optimization** **Credit: 3.00**

Introduction of the principal algorithms for linear, network, discrete, nonlinear, dynamic optimization and optimal control especially their methodology and the underlying mathematical structures. The simplex method, network flow methods, branch and bound and cutting plane methods for discrete optimization, optimality conditions for nonlinear optimization, interior point methods for convex optimization, Newton's method, heuristic methods, and dynamic programming and optimal control methods.

Recommended Texts:

1. E. K. P. Chong et al., *An Introduction to Optimization: With Applications to Machine Learning*, 5th ed. Wiley, 2023

CSE 4849 **Human Computer Interaction** **Credit: 3.00**

Foundations, The Human: Input-output channels, Human memory, Thinking: Reasoning and problem solving, individual Differences, Psychology and the Design of interactive Systems.

The Computer: Text Entry Devices, Output Devices, Memory, And Paper: Printing and scanning, processes.

The Interaction: Models of Interaction, Frameworks and HCI, Ergonomics, Interaction styles, The context of the Interaction.

Design Practice: Paradigms for interaction, Principles to support Usability, Using Design Rules, Usability Engineering, Interactive Design and Prototyping, Modules of the user in Design: Cognitive Models, Goal and Task Hierarchies, Linguistic Models. The challenges of Display Based Systems, cognitive Architectures; Task Analysis: Task Decomposition, Knowledge Based Analysis, E-R Based Techniques, Sources Information and Data Collection, Uses of Task Analysis. Dialogues Notations and Design: Dialogue Notations, Textual Dialogue Notations, Dialogue Semantics, Dialogue Analysis and Design; Models of the System: Standard Formalisms, Interaction Models, Status/Event